IA Mathematics for Natural Sciences: Examples Sheet 0: Basic Skills

Model Solutions

Please send all comments and corrections to jmm232@cam.ac.uk.

Making friends

1. Speak to your supervision partner, and, if you don't know already, find out: (a) where they are from; (b) their favourite food; (c) what they like to do to relax; (d) what part of the maths course they are most excited about this year. [*If you have the misfortune of being a singleton, ask someone else in your year instead!*]

> **Solution:** I'll tell you my answers, for fun:
>
> (a) I'm from a town called Lowestoft, in Suffolk. It's the most easterly place in Britain, and has a rather nice beach!
>
> (b) It's hard to say, but I do love trifle as a dessert.
>
> (c) I like playing piano, reading (mostly science-fiction), cooking and baking, and sometimes running.
>
> (d) The part of the first-year maths course that I like the most is the limits and continuity topic in Maths B - it's unusual to see this in a science course, and I think it helps to build an appreciation of what more formal mathematics is about.
>
> So, what was the point of this question? Like all of your courses this year, the maths course is *hard*. It is worth making some friends who are doing the course, so you can discuss the problem sheets together. If you can't do a question, ask someone else in your cohort if they have done it. If someone else in your cohort asks you how to do a problem, try to give them some help if you have time. You should treat the other Natural Scientists (and Chemical Engineers, and Computer Scientists) as team members - try to work together and help each other through the year!

Basic logic

2. Explain the meaning of the logical symbols $\Rightarrow$, $\Leftarrow$ and $\Leftrightarrow$. [*If you haven't seen them before, look them up online! In general, you should feel free to look up terms you don't understand on an examples sheet.*] Decide which of the following are true:

$$\text{(a) } x^2 \leq 1 \;\Rightarrow\; x \leq 1, \qquad \text{(b) } x^2 \leq 1 \;\Leftarrow\; x \leq 1, \qquad \text{(c) } x^2 \leq 1 \;\Leftrightarrow\; x \leq 1.$$

Similarly, decide which of the following are true:

$$\text{(d) } |x| = 1 \text{ if } x = 1, \qquad \text{(e) } |x| = 1 \text{ only if } x = 1, \qquad \text{(f) } |x| = 1 \text{ if and only if } x = 1.$$

Solution: The symbol $\Rightarrow$ means 'implies'. If we write $A \Rightarrow B$, then it means that 'if statement A is true, then statement B is true'. Similarly, the symbol $\Leftarrow$ means 'is implied by'. If we write $A \Leftarrow B$, then it means that 'if statement B is true, then statement A is true'. Finally, the symbol $\Leftrightarrow$ means 'implies and is implied by'. If we write $A \Leftrightarrow B$, then it means 'if statement A is true, then B is true, and vice-versa'.

Examining the given statements:

(a) The first statement is telling us that if $x^2 \leq 1$, then $x \leq 1$. This is true: the quadratic $x^2 - 1$ is less than or equal to zero in the region $-1 \leq x \leq 1$, which implies that $x \leq 1$ as required.

(b) The second statement is telling us that if $x \leq 1$, then $x^2 \leq 1$. This isn't true: if $x = -2$, then $x^2 = 4$, which is not less than or equal to one.

(c) The third statement is telling us that if $x \leq 1$, then $x^2 \leq 1$, and vice-versa. This isn't true, because of the implication in part (b).

In the second half of the question, the logical implications are written with words:

- A if B is the same as $A \Leftarrow B$;
- A only if B is the same as $A \Rightarrow B$ (that is, A cannot be true unless B is also true, hence the 'only if');
- A if and only if B is the same as $A \Leftrightarrow B$.

Examining the second set of given statements then, we have:

(d) If $x = 1$, then it is true that $|x| = 1$, so this statement is true.

(e) It is not true that $|x| = 1$ holds only if $x = 1$. For example, if $x = -1$, then we also have $|x| = 1$.

(f) The last statement cannot hold because of the implication in part (e).

3. What is the error in the following argument?

   'Suppose we wish to solve the equation $x - 1 = 2$. We begin by squaring both sides, to obtain $(x-1)^2 = 4$. Expanding the left hand side, we have:

   $$x^2 - 2x + 1 = 4 \qquad \Leftrightarrow \qquad x^2 - 2x - 3 = 0 \qquad \Leftrightarrow \qquad (x+1)(x-3) = 0.$$

   Since one of these factors must be zero, it follows that there are two solutions to the equation, $x = -1$ or $x = 3$.'

   Solution: The problem lies in the very first step, where the implication only goes in one direction! Writing the whole argument in logical symbols, we have:

   $$\begin{aligned} x - 1 = 2 \qquad &\Rightarrow \qquad (x-1)^2 = 4 \\ &\Leftrightarrow \qquad x^2 - 2x + 1 = 4 \\ &\Leftrightarrow \qquad x^2 - 2x - 3 = 0 \\ &\Leftrightarrow \qquad (x+1)(x-3) = 0 \\ &\Leftrightarrow x = -1, \text{ or } x = 3. \end{aligned}$$

   The logical symbol in the first line does *not* reverse. This shows that *if* the equation holds, *then* x must take one of the values in the final line. We have *not* shown that *if* x takes on of the values in the final line, *then* the equation is satisfied.

   Importantly, this shows us that when we are solving an equation, if we make a non-reversible logical step in our argument, we need to check that our solutions satisfy the original equation.

4. What is meant by *proof by contradiction*? Prove by contradiction that $\sqrt{2}$ is irrational.

   Solution: A 'proof by contradiction' is a sequence of logical steps $A \Rightarrow B \Rightarrow C \Rightarrow ... \Rightarrow Z$ where Z is *false*. This implies that the starting point, A, must be false too!

   Let's use proof by contradiction to show that $\sqrt{2}$ is irrational. We suppose that $\sqrt{2}$ is rational, so that we can write $\sqrt{2} = p/q$ for some integers p, q. We may assume that the fraction p/q is written in its lowest terms, so that p, q are not both even. Then squaring both sides, we obtain:

   $$2 = p^2/q^2 \qquad \Rightarrow \qquad 2q^2 = p^2.$$

   Now the left hand side is even, so p^2 must be even. Thus, p must be even, because the square of an odd number is odd. Thus, we can write $p = 2r$ for some integer r. Substituting into the above equation, we have:

   $$2q^2 = 4r^2 \qquad \Rightarrow \qquad q^2 = 2r^2.$$

   But then, by the same argument, q must be even. This contradicts our original assumption that p/q was written in its lowest terms. Hence $\sqrt{2}$ is irrational.

Sets and functions

5. State what is meant by the sets $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$. Decide which of the following statements are true:

$$\text{(a) } \pi \in \mathbb{Q}, \qquad \text{(b) } 3 \notin \mathbb{R}, \qquad \text{(c) } \mathbb{Z} \subseteq \mathbb{Q}, \qquad \text{(d) } \mathbb{Q} \supset \mathbb{C}.$$

> **Solution:** The given sets have the following definitions:
>
> - The set $\mathbb{Z}$ is the set of all integers, both positive and negative: $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$.
> - The set $\mathbb{Q}$ is the set of all rational numbers, i.e. numbers which are the ratio of two integers, i.e. numbers which are expressible as fractions.
> - The set $\mathbb{R}$ is the set of all real numbers, i.e. it includes both fractions, and numbers with an infinite, non-repeating decimal expansion.
> - The set $\mathbb{C}$ is the set of all complex numbers, i.e. numbers of the form $x + iy$ where x, y are real numbers. We will study complex numbers a lot more later in Michaelmas term!
>
> The symbol $\in$ means 'is a member of the set', and the symbol $\notin$ means 'is not a member of the set'. The symbol $\subseteq$ means 'is a subset of, or is equal to, the set'. The symbol $\supset$ means 'is a superset of'. Hence:
>
> (a) $\pi \in \mathbb{Q}$ is false, because π is not rational - it can't be written as a fraction!
>
> (b) $3 \notin \mathbb{R}$ is false, because 3 is a real number.
>
> (c) $\mathbb{Z} \subseteq \mathbb{Q}$ is true, because every integer is also a rational number (we can write the integer p as a fraction $p/1$, if you are worried!).
>
> (d) $\mathbb{Q} \supset \mathbb{C}$ is false, because not every complex number if a ratio of integers. For example, i is not a ratio of two integers!

6. State what is meant by the sets $[a, b]$, (a, b), and $(a, b]$, where a, b are real numbers. Decide which of the following statements are true:

$$\text{(a) } 1 \in [0, 1), \qquad \text{(b) } 3 \notin (3, 4), \qquad \text{(c) } [2, 3] \subset (2, 5], \qquad \text{(d) } (-1, 0) \subset [-1, 0].$$

> **Solution:** These sets are *intervals* of real numbers. They have the following definitions:
>
> - $[a, b]$ is the set of all real numbers x in the range $a \leq x \leq b$, so it *includes* the endpoints.
> - (a, b) is the set of all real numbers x in the range $a < x < b$, so it *excludes* the endpoints.
> - $(a, b]$ is the set of all real numbers x in the range $a < x \leq b$, so it excludes the endpoint a, but includes the endpoint b.
>
> Out of the given statements, we have:
>
> (a) $1 \in [0, 1)$ is false, because the curved bracket means that the endpoint of the interval is excluded.
>
> (b) $3 \notin (3, 4)$ is true, because the endpoint 3 is excluded from the interval in this case.
>
> (c) $[2, 3] \subset (2, 5]$ is false, because 2 is in $[2, 3]$ but not in $(2, 5]$.
>
> (d) $(-1, 0) \subset [-1, 0]$ is true, because every real number satisfying $-1 < x < 0$ also satisfies $-1 \leq x \leq 0$.

7. A function is a mapping from a set to another set. We write $f : A \to B$ to denote the function (or just f when the sets are implied), and we write $f(x)$ for the value of the function at the point $x \in A$. Which of the following define functions, and why?

   (a) $f : [0, \infty) \to \mathbb{R}$ given by $f(x) = \sqrt{x}$;

   (b) $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(x) = \frac{1}{2}x$;

   (c) $f : \mathbb{R} \to \mathbb{R}$ where $f(x)$ is implicitly defined by $2^{f(x)} = x$;

   (d) $f : [0, \infty) \to \mathbb{R}$ where $f(x)$ is implicitly defined by $f(x)^2 = x$.

   Solution:

   (a) This defines a function, because for every value of $x \in [0, \infty)$, the function uniquely gives a value $\sqrt{x} \in \mathbb{R}$. Note that we don't hit *every* real number with this function, because the square root is always positive - that doesn't matter though, as long as the result is a member of the set of real numbers. The set of things we hit is called the *range* (or *image*) of the function.

   (b) This does *not* define a function, because if $x = 1$, then $f(1) = 1/2$ is not in the set that the function maps to, which is supposed to be the integers. We could fix this by saying that the function is a mapping $f : \mathbb{Z} \to \mathbb{Q}$ or $f : \mathbb{Z} \to \mathbb{R}$.

   (c) This does not define a function, because for $x = -1$ for example, there is no real number $f(x)$ such that $2^{f(x)} = -1$. This is because exponentiation of a positive real number always gives a positive result. This could be fixed by considering the function to be a mapping $f : (0, \infty) \to \mathbb{R}$; the function that this defines is then just $f(x) = \log_2(x)$.

   (d) This does define a function, because it is multi-valued. For example, take $x = 1$. Then $f(x)^2 = 1$ has two roots, $f(x) = \pm 1$. We don't know which one to pick! If we instead said the function was a mapping $f : [0, \infty) \to [0, \infty)$, we would know to pick the positive root, which ten makes the function equivalent to the one in part (a).

Sequences and series

8. Prove the formula:
$$\sum_{k=1}^{n} k = \tfrac{1}{2}n(n+1)$$
by (a) induction; (b) considering the sum of $(k+1)^2 - k^2$ from $k = 1$ to $k = n$. Is it more useful to prove a result by induction, or by a direct argument? Why?

Solution: For the inductive argument, the base case $n = 1$ is obvious since:
$$\sum_{k=1}^{1} k = 1, \qquad \frac{1(1+1)}{2} = 1.$$
Now assume the result is true for $n = m$, and consider the case $n = m + 1$. We have:
$$\begin{aligned}
\sum_{k=1}^{m+1} k &= (m+1) + \sum_{k=1}^{m} k \\
&= (m+1) + \frac{m(m+1)}{2} && \text{(induction hypothesis)} \\
&= (m+1)\left(1 + \frac{m}{2}\right) \\
&= \frac{(m+1)(m+2)}{2}.
\end{aligned}$$
Hence the result is true for $n = m + 1$, if the induction hypothesis holds. So we're done by the principle of mathematical induction.

To prove the result by considering the sum of $(k+1)^2 - k^2$, we first note that the resulting sum *telescopes*:
$$\sum_{k=1}^{n}\left((k+1)^2 - k^2\right) = (2^2 - 1^2) + (3^2 - 2^2) + (4^2 - 3^2) + (5^2 - 4^2) + \cdots + (n+1)^2 - n^2 = (n+1)^2 - 1.$$
We are able to write down the sum, because there is cancellation between any two consecutive terms. On the other hand, we also have $(k+1)^2 - k^2 = 2k + 1$. Therefore, we have shown that:
$$\sum_{k=1}^{n}(2k+1) = (n+1)^2 - 1 = n^2 + 2n. \qquad (*)$$
Using standard properties of summation, we can split the sum on the left and take out the overall factor, leaving:
$$\sum_{k=1}^{n}(2k+1) = 2\sum_{k=1}^{n} k + \sum_{k=1}^{n} 1 = 2\sum_{k=1}^{n} k + n.$$
Rearranging and using the above equality $(*)$, we obtain the required result:
$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

> Induction is great whenever we *know the result in advance* (that is, we know what we want to prove!). If we were trying to find the sum of k from $k = 1$ to n in the first place, the telescopic series method would be more useful. Indeed, it generalises well to $(k+1)^3 - k^3 = 3k^2 + 3k + 1$, allowing us to find the sum of the squares k^2 given the sum of k already. With induction, we would need a 'guess' for a formula first.

9. What is meant by an *arithmetic sequence* with first term a and common difference d? Prove that the sum of the first n terms of an arithmetic sequence is

$$S_n = \tfrac{1}{2}n(2a + (n-1)d).$$

Hence, find the sum of the series $2, 5, 8, 11, ..., 32$.

> **Solution:** The arithmetic sequence with first term a and common difference d is:
>
> $$a, \qquad a+d, \qquad a+2d, \qquad a+3d, \qquad ...$$
>
> The sum of the first n terms is given by:
>
> $$\sum_{k=0}^{n-1}(a+kd) = na + d\sum_{k=0}^{n-1}k = na + \frac{d(n-1)n}{2} = \frac{1}{2}n(2a+(n-1)d),$$
>
> as required. A particularly easy way to remember this is to notice it is the average of the first and last terms in the series, multiplied by the number of terms in the series:
>
> $$\frac{1}{2}n(2a+(n-1)d) = n\cdot\frac{1}{2}\left(a + (a+(n-1)d)\right).$$
>
> The given sequence $2, 5, 8, 11, ..., 32$ is an arithmetic sequence with first term $a = 2$ and common difference $d = 3$. The final term is the $(32-2)/3 + 1 = 11$th term in the series. Hence the sum is:
>
> $$\frac{11}{2}(4 + 10\cdot 3) = \frac{11\cdot 34}{2} = 11\cdot 17 = 187.$$

10. What is meant by a *geometric sequence* with first term a and common ratio r? Prove that if $r \neq 1$, the sum of the first n terms of a geometric sequence is:

$$S_n = \frac{a(1-r^n)}{1-r}.$$

What happens if $r = 1$? What is the behaviour of the sum in the limit as $n \to \infty$? Hence, find the sum of the infinite series $2, 2/3, 2/9, 2/27, ...$

> **Solution:** The geometric series with first term a and common ratio r is:
>
> $$a, \qquad ar, \qquad ar^2, \qquad ar^3, \qquad ar^4, \qquad ...$$
>
> The sum of the first n terms is:
>
> $$S_n = a + ar + ... + ar^{n-1}.$$
>
> Multiplying by $1-r$, we have:
>
> $$S_n(1-r) = (a + ar + ... + ar^{n-1})(1-r) = a + ar + ... + ar^{n-1} - ar - ar^2 - ... - ar^n = a - ar^n.$$

Hence, the sum is indeed given by:

$$S_n = \frac{a(1-r^n)}{1-r},$$

as required. Observe that this formula *only* applies if $r \neq 1$; in the case where $r = 1$, the sum to n terms is just na (it's just $a + a + ... + a$, but n times). Whenever we have a sequence where we are dividing by zero in maths in some special case, we should return to the first place where we tried to divide by zero and see what happens if we are in that special case.

If $|r| < 1$, we have $r^n \to 0$ as $n \to \infty$. Hence if $|r| < 1$, we can sum a geometric series to infinity, giving the result:

$$S_\infty = \frac{a}{1-r}.$$

In the case of the series $2, 2/3, 2/9, 2/27, ...$ the first term is $a = 2$ and the common ratio is $r = 1/3$, so the sum to infinity is:

$$S_\infty = \frac{2}{1-1/3} = \frac{2}{2/3} = 3.$$

11. Using the formula for the sum of an infinite geometric series, find a formula for the sum of the infinite series:

$$\sum_{k=1}^{\infty} kr^k,$$

where $|r| < 1$. Hence, determine $2/3 + 2\,(2/3)^2 + 3\,(2/3)^3 + 4\,(2/3)^4 + \cdots$.

Solution: In the previous question, we prove that:

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}.$$

Multiplying through by r, we have:

$$\sum_{k=1}^{\infty} ar^k = \frac{ar}{1-r}.$$

Now comes the clever step. If we *differentiate* both sides of the formula with respect to r, we have:

$$\sum_{k=1}^{\infty} akr^{k-1} = \frac{d}{dr}\left(\frac{ar}{1-r}\right) = \frac{a(1-r)+ar}{(1-r)^2} = \frac{a}{(1-r)^2}.$$

To finish, we set $a = 1$ and multiply through by r, giving:

$$\sum_{k=1}^{\infty} kr^k = \frac{r}{(1-r)^2}.$$

For the given series, we have the value:

$$\sum_{k=1}^{\infty} k\left(\frac{2}{3}\right)^k = \frac{2/3}{(1-2/3)^2} = \frac{2/3}{1/9} = 6.$$

Trigonometric functions and their identities

12. Define the trigonometric functions $\sin(x), \cos(x), \tan(x)$ using the sides of a right-angled triangle. Define also the reciprocal trigonometric functions $\operatorname{cosec}(x)$, $\sec(x)$, $\cot(x)$. Hence, prove each of the following trigonometric identities:

    (a) *The Pythagorean identities:* $\sin^2(x) + \cos^2(x) = 1$, $\tan^2(x) + 1 = \sec^2(x)$, $\cot^2(x) + 1 = \operatorname{cosec}^2(x)$.

    (b) *The cofunction identities:* $\sin(\frac{\pi}{2} - x) = \cos(x)$, $\cos(\frac{\pi}{2} - x) = \sin(x)$, $\tan(\frac{\pi}{2} - x) = \cot(x)$.

    (c) *The compound angle formulae:*

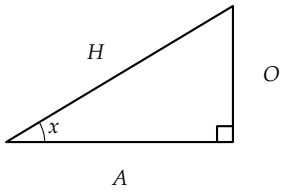
$$\sin(x \pm y) = \sin(x)\cos(y) \pm \sin(y)\cos(x), \quad \cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y), \quad \tan(x \pm y) = \frac{\tan(x) \pm \tan(y)}{1 \mp \tan(x)\tan(y)}.$$

    (d) *The double angle formulae:*

$$\sin(2x) = 2\sin(x)\cos(x), \qquad \cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 = 1 - 2\sin^2(x).$$

    (e) *The power reduction formulae:* $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$, $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$.

    Learn all of these identities off by heart, and get your supervision partner to test you on them.

> **Solution:** There are lots of ways to define the trigonometric functions $\sin(x), \cos(x), \tan(x)$. For example, these functions can be defined in terms of complex exponentials, Taylor series, or as solution of certain differential equations (we shall study all of these options later in the course). Here, however, we will use the most elementary definition in terms of right-angled triangles.
>
> 
>
>
> Given a right-angled triangle with sides O, A, H, and one given angle x, where H is the length of the hypotenuse and O is the length of the sides opposite the angle x, we shall define the trigonometric functions via:
>
> $$\sin(x) := \frac{O}{H}, \qquad \cos(x) := \frac{A}{H}, \qquad \tan(x) := \frac{O}{A} = \frac{O}{H} \cdot \frac{H}{A} = \frac{\sin(x)}{\cos(x)}.$$
>
> The *reciprocal trigonometric functions* are then defined as:
>
> $$\operatorname{cosec}(x) := \frac{1}{\sin(x)}, \qquad \sec(x) := \frac{1}{\cos(x)}, \qquad \cot(x) := \frac{1}{\tan(x)}.$$
>
> Notice that this definition *only* works for x in the range $x \in [0, \pi/2)$, because we need it to be an acute angle. We extend the definition by requiring the standard periodicity properties $\sin(x) = \sin(\pi - x)$ (extending to $[0, \pi]$), $\sin(-x) = -\sin(x)$ (further extending to $[-\pi, \pi]$) and $\sin(x) = \sin(x + 2\pi)$ (extending to all real numbers); similarly $\cos(x) = -\cos(\pi - x)$ (extending to $[0, \pi]$), $\cos(x) = \cos(-x)$ (extending to $[-\pi, \pi]$), and $\cos(x) = \cos(x + 2\pi)$ (extending to all real numbers). We can see *why* these extensions are used by putting the triangle inside a unit circle ($H = 1$), and then taking A, O to be the cosine and sine respectively as the angle x to the x-axis is varied in the full range.

We can now immediately prove the given identities:

(a) *The Pythagorean identities.* These identities are called the Pythagorean identities because they come from Pythagoras' theorem for the right-angled triangle: $O^2 + A^2 = H^2$. Dividing by H^2, we have:

$$\left(\frac{O}{H}\right)^2 + \left(\frac{A}{H}\right)^2 = 1 \qquad \Leftrightarrow \qquad \sin^2(x) + \cos^2(x) = 1.$$

Dividing this identity by $\cos^2(x)$, we have:

$$\frac{\sin^2(x)}{\cos^2(x)} + 1 = \frac{1}{\cos^2(x)} \qquad \Leftrightarrow \qquad \tan^2(x) + 1 = \sec^2(x).$$

On the other hand, we could have divided instead by $\sin^2(x)$, which would have resulted in:

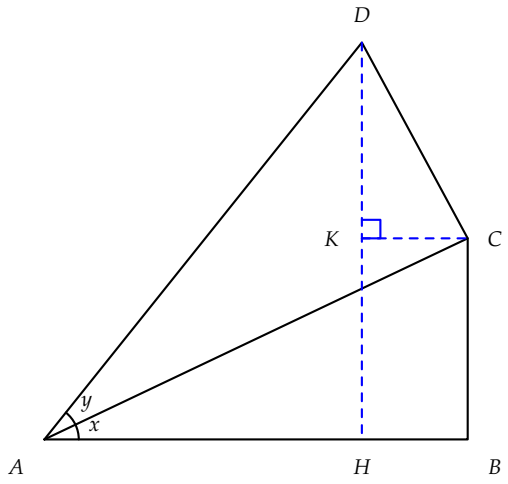
$$1 + \frac{\cos^2(x)}{\sin^2(x)} = \frac{1}{\sin^2(x)} \qquad \Leftrightarrow \qquad 1 + \cot^2(x) = \operatorname{cosec}^2(x),$$

which completes our derivation of the three Pythagorean identities.

(b) *The cofunction identities.* These follow immediately from the right-angled triangle definition, because the remaining angle in a right-angled triangle (after we have specified one as x) is $\pi/2 - x$. Changing our angle swaps our perspective on what is the 'adjacent' side and what is the 'opposite' side. This gives immediately:

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x), \qquad \cos\left(\frac{\pi}{2} - x\right) = \sin(x), \qquad \tan\left(\frac{\pi}{2} - x\right) = \cot(x).$$

(c) *The compound angle formulae.* Consider the following diagram, involving two right-angled triangles ABC and ACD stacked on top of one another. One has an acute angle x at the vertex A, and one has an acute angle y at the vertex A (note that we assume that $x + y$ is acute).



Observe that x and the angle KCA are alternating, so KCA is equal to x. This implies that the triangles DKC and ABC are similar triangles.

Now, by the definition of sine, we have (writing AB for the length of the line segment from A to B, etc):

$$\begin{aligned}
\sin(x+y) &= \frac{DH}{AD} \\
&= \frac{DK}{AD} + \frac{KH}{AD} && \text{(since } DH = DK + KH\text{)} \\
&= \frac{DK}{AD} + \frac{CB}{AD} && \text{(since } KH = CB\text{)} \\
&= \frac{DK}{DC}\frac{DC}{AD} + \frac{CB}{AC} \cdot \frac{AC}{AD} \\
&= \cos(x)\sin(y) + \sin(x)\cos(y).
\end{aligned}$$

This establishes the formula in the case where $x+y$, x, y are acute; the formula is readily extended to all real values of the angle via the various properties we outlined above. In particular, this implies that we can replace $y \mapsto -y$, which gives:

$$\sin(x-y) = \sin(x)\cos(y) - \cos(x)\sin(y),$$

which gives the full formula $\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y)$. We can also replace $x \mapsto \pi/2 - x$ in the formula giving:

$$\sin\left(\frac{\pi}{2} - x \pm y\right) = \sin\left(\frac{\pi}{2} - x\right)\cos(y) \pm \cos\left(\frac{\pi}{2} - x\right)\sin(y).$$

Using the cofunction identities, this implies:

$$\cos(x \mp y) = \cos(x)\cos(y) \pm \sin(x)\sin(y),$$

as required. Finally, dividing the formula for $\sin(x \pm y)$ by the formula for $\cos(x \pm y)$, we obtain the tangent compound angle formula:

$$\tan(x \pm y) = \frac{\sin(x \pm y)}{\cos(x \pm y)} = \frac{\sin(x)\cos(y) \pm \cos(x)\sin(y)}{\cos(x)\cos(y) \mp \sin(x)\sin(y)} = \frac{\tan(x) \pm \tan(y)}{1 \mp \tan(x)\tan(y)},$$

where in the final step we divided the numerator and denominator by $\cos(x)\cos(y)$.

(d) *The double angle formulae.* These follow immediately from the compound angle formulae by taking $x = y$. We have:

$$\sin(2x) = \sin(x+x) = \sin(x)\cos(x) + \cos(x)\sin(x) = 2\sin(x)\cos(x),$$

and similarly:

$$\cos(2x) = \cos(x+x) = \cos^2(x) - \sin^2(x).$$

To get the alternative forms of the cosine double angle formulae, we use one of the Pythagorean identities in the following forms: $\cos^2(x) = 1 - \sin^2(x)$ and $\sin^2(x) = 1 - \cos^2(x)$. This gives:

$$\begin{aligned}
\cos(2x) &= \cos^2(x) - \sin^2(x) = (1 - \sin^2(x)) - \sin^2(x) = 1 - 2\sin^2(x), \\
\cos(2x) &= \cos^2(x) - \sin^2(x) = \cos^2(x) - (1 - \cos^2(x)) = 2\cos^2(x) - 1.
\end{aligned}$$

(e) *The power reduction formulae.* These follow immediately from the double angle formulae for $\cos(2x)$, simply by rearrangement:

$$\cos(2x) = 1 - 2\sin^2(x) \qquad \Rightarrow \qquad \sin^2(x) = \frac{1}{2}(1 - \cos(2x)),$$

$$\cos(2x) = 2\cos^2(x) - 1 \qquad \Rightarrow \qquad \cos^2(x) = \frac{1}{2}(1 + \cos(2x)).$$

Exams and coursework

13. (a) Look up the format of the first-year maths exams. How much is each exam worth? How long does each exam last? How many questions do you have to answer, and how should you split your time in the exams? Are you allowed a calculator? Is there a formula book?

(b) Look up the first-year maths coursework. How much is it worth?

Solution:

(a) There are two first-year maths exams, which are each worth $6/13 \approx 46\%$ of your final grade in maths (accounting for a total of $12/13 \approx 92\%$ of your final grade). Each exam lasts three hours. Each exam has the following structure:

- **Section A.** The exam begins with 10 short questions, each worth 2 marks, totalling 20 marks. These questions are designed to be answered quickly and credit is always awarded to the correct answer, even if full reasoning is not given. This section is compulsory.
- **Section B.** The second part of the exam contains 10 long questions, which are each worth 20 marks. You must choose **5 questions** of the **10 questions** to answer. These questions usually award marks for explaining yourself well, in addition to getting the right answers. Out of the 10 questions, 2 questions are only accessible to students who have done the Maths B course, but these will be clearly marked to avoid confusion.

In total, the exam is out of 120 marks, with 20 marks coming from Section A and 5 questions contributing 20 marks each in Section B. This means you should spend at most 30 minutes on Section A, and at most 30 minutes on each Section B question.

(b) The first-year maths coursework is a scientific computing course that is run by the Cavendish Laboratory (**not** by the maths department). It involves 8 practical sessions, each 90 minutes long, 4 of which you attend in Michaelmas and 4 of which you attend in Lent. Each practical involves submission of a small piece of computing work which contributes to your final grade. The total contribution is $1/13 \approx 6\%$ of your grade, so that each piece of work is worth $1/104 \approx 1\%$ of your final grade.